LIA ERISSON

+1(914) 705-3662 ◇ Boston, MA ◇ liaerisson.github.io
lia.erisson15@gmail.com ◇ linkedin.com/in/lia-erisson ◇ github.com/liaerisson

EDUCATION

B.A. in Computer Science & Honours Economics, McGill University 2024 – 2028
Relevant Coursework: C++, Computer Systems, Software Systems, Probability; Current: Data Structures & Algorithms, Programming Paradigms, Honours Econometrics, Macroeconomics.

TECHNICAL EXPERIENCE

Product Analyst Intern Jul 2026 – Aug 2026
CHEQ *(Cybersecurity)*

- Built a semantic knowledge base and SQL retrieval queries for a client-facing AI agent using data and relationships across three databases processing 1B+ daily site interactions for 15,000+ customers.
- Mapped undocumented data relationships and reconciled conflicting business definitions; tested outputs for retrieval errors, backend-logic issues, and restricted-data exposure with the AI team.

Rotational Technology Intern Jun 2026 – Jul 2026
I-NEXT Technology, Sourasky Medical Center

- Built and deployed AWS agent infrastructure for a clinician-scheduling prototype using Bedrock AgentCore and Docker, adapting the deployment upon identifying reliability issues and supporting an end-to-end hackathon demo.

Research Assistant Sep 2022 – Apr 2024
Prof. Hanna Lerner, Tel Aviv University

- Researched post-COVID labor-rights changes across global medical-goods supply chains, comparing academic and policy sources to identify patterns, inconsistencies, and open research questions.

PROJECTS

C++ Limit Order Book & Matching Engine *C++*

- Implemented order entry, cancellation, trader validation, and execution of crossing orders using price-time priority; processed up to 1.82 million orders/second in single-thread benchmark testing.
- Structured ordered bid/ask price levels, FIFO queues, and direct order-ID lookup using STL containers; built 79 Catch2 tests covering book operations, validation, and matching behavior.

Java Search Engine & REST API *Java, Spring Boot, JUnit*

- Developed document ingestion, inverted indexing, ranked multi-term retrieval, and a Spring Boot REST API for search and query access.
- Built tokenization, query-processing, and ranking logic with 31 JUnit tests for scoring, retrieval, and query behavior.

ACTIVITIES & LEADERSHIP

Martlet Fund McGill Sep 2026 – Present
Quantitative Trading Senior Analyst

- Research and implement quantitative trading strategies for the fund's paper-trading framework; evaluate strategies and market conditions through literature review and statistical analysis.
- Mentor two junior analysts through strategy development, implementation, and testing.

McGill Women in Computer Science Mar 2025 – Present
VP Events (Jul 2026 – Present) | Events Director (Mar 2025 – Jun 2026)

- Lead a 7-person team organizing 1–2 events monthly and coordinate with 20+ company, faculty, and student partners; host multiple events with 100+ attendees annually.

SKILLS

Programming: Java, Python, C++, SQL, C, Bash, OCaml
Cloud & Tools: AWS, Amazon Bedrock, Docker, Git, Linux, CMake, Maven
Backend & Data: Spring Boot, REST APIs, Relational Databases, Information Retrieval, Unit Testing, OOP